



Society of Petroleum Engineers

SPE-227241-MS

Ultrasound Well Stimulation in Context: A Comparative Review with Other Well Stimulation Techniques

Rima T. Alfaraj and Murtadha J. AlTammar, Saudi Aramco, Dhahran, Saudi Arabia; Jesus Mierez, Rexonic Ultrasonics AG, Wil, Switzerland

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227241-MS](https://doi.org/10.2118/227241-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a comparative review of well stimulation tools, with a particular focus on comparing high-power ultrasound to other relevant mechanical stimulation tools including dynamic fluid pulsing, low-frequency acoustic stimulation, plasma pulse, and pressure wave systems. The evaluation positions ultrasound stimulation within the broader landscape of stimulation technologies by comparing each method's mechanical operation, performance, depth of treatment, and compatibility with common chemical stimulation methods. While each stimulation technique offers a unique energy delivery mechanism and treatment characteristics, ultrasound stimulation offers a balanced, efficient, and low-risk option. This evaluation highlights each method's distinct operational profile, enabling more informed and sustainable stimulation strategy selection based on well and reservoir-specific conditions.

Keywords: Ultrasonic stimulation, Well stimulation, Oil recovery, Comparative review, High-power ultrasound

Introduction

The stimulation of oil and gas wells is a critical process aimed at enhancing reservoir productivity by improving permeability and addressing formation damage. As the oil and gas industry explores alternatives to traditional chemical stimulation, various mechanical methods have emerged, including high-power ultrasound (US) tools. Several recent publications have presented experimental and laboratory-scale findings related to ultrasound stimulation. Mierez et al. (2024) [1] provided a review outlining current ultrasound applications across different oilfield operations, including well stimulation. Alfaraj et al. (2024) [2] reported laboratory evaluations demonstrating the impact of ultrasound tools on flow improvement in core samples. Additionally, AlTammar et al. (2025) [3] conducted yard-scale experiments to assess the tool's effectiveness in stimulating tight formations. These studies contribute to the broader body of research investigating mechanical stimulation tools, offering a reference point for comparison with other tools explored in this review.

In the pursuit of improving well performance, it is crucial to constantly assess and compare all available technologies and recognize their potential for real-world field operations. By conducting a comparative analysis of ultrasonic stimulation technology against other commonly known mechanical stimulation tools, it was possible to establish a foundational framework for a thorough and systematic comparison, including the optimization of key operational parameters, and identification of critical research gaps that could yield valuable insights and support for the implementation of more sustainable and efficient technologies in the industry.

This paper provides a comparative analysis of ultrasound stimulation and various mechanical stimulation tools. The review explores technologies such as dynamic fluid pulsing, low-frequency acoustic stimulation, plasma pulse, and pressure wave systems. While some studies have examined these techniques individually, few have conducted a structured comparison to highlight their respective strengths, limitations, and areas of potential integration. To address this gap, the paper compiles and analyzes existing literature—peer-reviewed articles, technical papers, and industry reports—to establish a clear framework for evaluating these stimulation methods. By framing ultrasound stimulation within the broader landscape of mechanical well stimulation technologies, this study aims to offer insights into how ultrasound tools perform relative to other innovative methods, supporting more informed decision-making in future field implementations. Additionally, this comparative analysis can guide the development of hybrid approaches, where the synergistic combination of ultrasonic stimulation and other technologies may lead to further enhancements in hydrocarbons recovery efficiency.

For the comparison of the US against other stimulation tools, personalized criteria are considered for each to comprehensively evaluate them in equivalent conditions mainly due to the inexistence of sufficient comparative research, theoretical or experimental. Dynamic fluid pulsing, seismic or low frequency acoustic stimulation, electro-hydraulic stimulation and plasma pulse were the most relevant unconventional technologies found in the literature for this contrast.

The study outlines the key features, operational principles, and application context of mechanical stimulation tools compared to ultrasonic stimulation, offering a structured overview of their use in enhancing well stimulation. The comparison is intended to document current stimulation approaches within the industry and to outline how ultrasound fits within the broader landscape of stimulation tools.

Dynamic Fluid Pulsing (DFP)

Dynamic Fluid Pulsing (DFP) is a mechanical well stimulation technique that has gained attention for its ability to enhance fluid flow and permeability through pressure pulse-driven dispersion. The foundational research on this method dates back to 1967, when Felsenthal and Ferrel [8] first presented pressure pulsing as a promising stimulation mechanism. The modern DFP system was further developed in the late 1990s by Canadian researchers who designed it to drive fluid into formation areas beyond the path of least resistance. The mechanism involves the generation of instantaneous, high-frequency pressure pulses that create a rapid dispersion of fluid radially into the formation. This induces localized expansion and contraction of pore structures, enhancing dynamic permeability and improving fluid mobility.

DFP begins with wellbore cleaning using a cavitation-based tool that creates a controlled water hammer effect to remove scale and flow restrictions. The stimulation tool itself consists of a magnetically controlled piston assembly actuated by a pressure differential. The piston cyclically opens and closes to release high-velocity fluid pulses—up to 100 m/s—into the formation. This process significantly improves the penetration of treatment fluids and supports deeper, more uniform distribution. As shown in (Figure N° 1) DFP changes the fluid injection pattern by promoting radial flow into the reservoir matrix [9]. Figure N° 2 illustrates the improved fluid dispersion before and after DFP treatment [4].

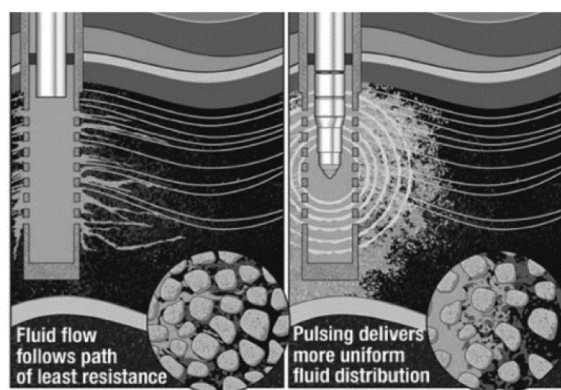


Figure 1—Fluid injection effect in the reservoir. On the left, standard injection; on the right, DFP injection [10].

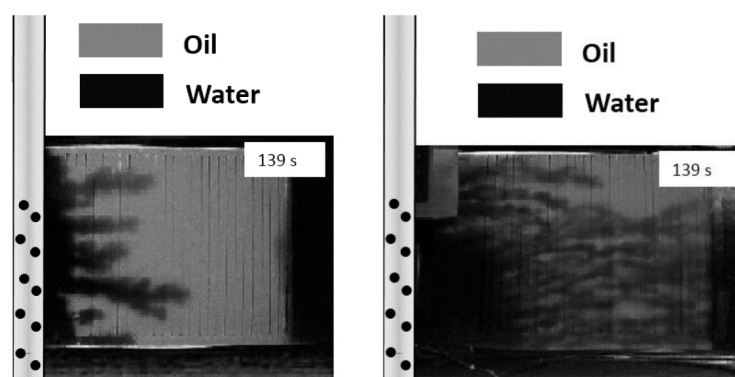


Figure 2—DFP dispersion model before and after treatment [5].

Operational specifications for DFP tools include a differential pressure range of 300–1,200 psi, with maximum temperature ratings of 110 °C (230 °F). The tools can withstand pressures up to 1,400 psi and are designed for continuous operation up to 72 hours. With an outer diameter (OD) of 1.69 inches, DFP tools are deployable using coiled tubing or jointed pipe, providing flexibility across different well geometries.

DFP's primary stimulation zone is near the wellbore, with an estimated penetration depth of up to 3 feet. Its effectiveness is influenced by factors such as fluid injection pressure, pulse amplitude, and reservoir rock properties. Despite its practical applications, the underlying mechanics of fluid pulse propagation and formation interaction remain poorly understood. No universal predictive models exist; however, dynamic simulation models are used for treatment optimization [11].

When compared to ultrasound (US) stimulation, several key differences and similarities arise. Both methods enhance permeability and reduce formation damage, but US stimulation supports higher operational pressures (up to 4,300 psi) and temperatures (up to 120 °C). US tools typically have a slightly larger OD (>1.75 in) and longer operational run times exceeding one week. Both technologies are compatible with multiple stimulation approaches, including chemical, water, and surfactant-based.

Applications overlap significantly between DFP and US stimulation, including wax and scale removal, formation damage mitigation, and enhanced injectivity. However, US stimulation is also effective for asphaltene removal and is deployable in production as well as injection wells. In contrast, DFP is more often applied in water injectors and for remedial stimulation in injection-focused operations.

Notably, field results demonstrate DFP's ability to increase oil recovery. In waterflooded reservoirs with permeability below 50 mD, recovery improvements of up to 52% have been observed [8]. Tan et al. (2023) [12] also reported that synchronizing pulsing frequencies with a reservoir's natural frequency induced resonance effects that enhanced oil mobility. Wu et al. (2025) [13] validated these findings in high water cut reservoirs, showing a 14% increase in recovery rates using hydraulic pulsation.

DFP provides an effective stimulation solution for near-wellbore enhancement. While ultrasound stimulation offers broader operational envelopes and additional mechanisms such as cavitation, DFP remains a valuable tool for specific applications, particularly where hydraulic pulsing provides a strategic advantage.

Seismic or Low-Frequency Acoustic Stimulation

Seismic or low-frequency acoustic stimulation is a mechanical well stimulation technique that employs low-frequency, high-energy elastic waves to improve oil mobility and recovery. This method was first explored in the 1950s in Russia and has since evolved as a cost-effective and environmentally conscious alternative to fluid or chemical-based stimulation.

The core mechanism involves a permanently installed downhole tool, often placed in non-producing or abandoned wells at depths ranging from 700 to 10,000 feet. The tool is powered by a conventional surface pumping unit, which drives a rod string connected to synchronized plungers. These plungers compress and release fluid in a repetitive cycle, producing elastic shock waves that radiate laterally and vertically through the reservoir [14]. These waves mobilize trapped oil droplets by breaking adhesive forces and overcoming capillary resistance.

Seismic waves can propagate unimpeded across faults, stratification, and changes in lithology, making this method particularly suitable for heterogeneous or compartmentalized reservoirs. A single device can stimulate wells within a radius of up to 1.4 miles. Each unit delivers an elastic wave every 10 seconds—translating to over 2 million waves during its estimated lifespan of 1.5 years. Importantly, the stress levels remain below the thresholds that would damage well casing or cement, maintaining wellbore integrity throughout the stimulation process.

Despite the theoretical advantages, field results have been mixed. A 2013 test in the Permian Basin near Andrews, Texas, intended for 8 months of continuous operation, was aborted after just one day due to diminished tool output [15]. This case underscores the importance of durability, real-time monitoring, and design improvements for successful long-term application.

Ultrasound (US) stimulation offers a more flexible and targeted approach, rather than long-term deployment. US is applied as a short-duration intervention directly in the target well. US stimulation generates high-frequency acoustic waves that induce cavitation and micro-fractures in the near-wellbore region, enhancing permeability with minimal formation damage. Its mobility, precision, and ability to operate in both continuous and pulsed modes make it suitable for a wide variety of well conditions and geometries.

US tools also allow integration with other stimulation methods, such as chemical treatments, which is not typical in seismic applications. While US stimulation offers localized but immediate enhancements, seismic methods act over extended periods with broader radial influence, making them less controllable but potentially more impactful in certain geologic settings. While seismic stimulation provides long-range effects and reservoir-wide influence, ultrasound stimulation offers precision, speed, and operational flexibility. Each method brings unique advantages depending on the reservoir characteristics and stimulation objectives.

Electro-Hydraulic Stimulation (EHS)

Electro-Hydraulic Stimulation (EHS) is a high-energy well stimulation method that transforms stored electrical energy into powerful hydraulic shock waves through a rapid plasma discharge. This energy, discharged within microseconds, superheats and vaporizes surrounding fluid to generate a high-pressure pulse that propagates through the reservoir formation [16–19]. These pulses can effectively dislodge formation damage, stimulate permeability, and induce fracturing near the wellbore.

EHS tools are conveyed using standard wireline and operate in fluid-filled wellbores, eliminating the need for specialized surface power units. The energy released during the plasma arc can reach several hundred megawatts, generating stress waves that radiate at the speed of sound (~ 1500 m/s). The induced stress exceeds the tensile strength of most reservoir rocks but remains below the yield strength of steel, ensuring casing and cement integrity under normal conditions.

The stimulation is focused in the near-wellbore zone, reaching approximately 0.9 meters (3 feet) into the formation. However, caution is needed to avoid unintended damage. Experiments by Liu et al. [17] demonstrated the effectiveness of EHS in generating fractures, but also revealed that excessive shock intensity could lead to damage in external casing and cement. Figure N° 3 illustrates post-treatment rock specimens, highlighting both the fracture propagation and risks associated with uncontrolled energy discharge.

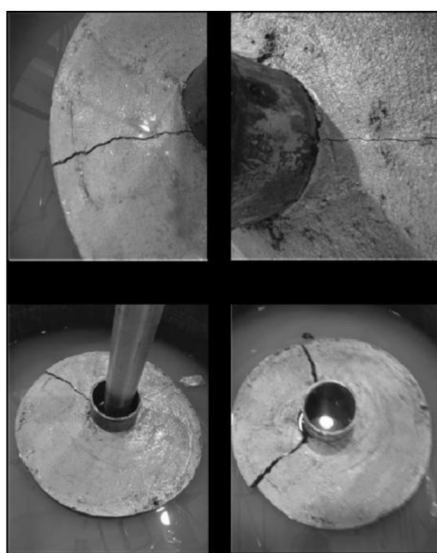


Figure 3—Pictures of specimen after EHS by Liu et al. [17].

Ultrasound (US) stimulation offers a more controlled, non-destructive mechanism through acoustic cavitation. US tools function at slightly lower temperatures (up to 120 °C versus 130 °C for EHS) but support higher pressures (up to 4,300 psi), providing better-defined operational safety limits. Tool dimensions also differ: EHS tools typically have a 2.125-inch OD, while US tools are slimmer at around 1.75 inches, allowing better access in tighter well completions. EHS is typically a standalone mechanical process that can be used before or after chemical stimulation to open flow paths, while US can be used in conjunction with chemical stimulation.

Both EHS and US technologies can be deployed in vertical, deviated, or horizontal wells using wireline, coiled tubing, or tractors. They are compatible with production and injection wells and require a fluid column for wave propagation. EHS stimulation is always pulsed, releasing energy in short, powerful bursts. In contrast, US stimulation supports both continuous and pulsed modes, giving operators more flexibility in tailoring treatment to reservoir conditions.

Application-wise, the two methods are similar: both are used for enhancing permeability, reducing formation damage, and removing paraffin, wax, asphaltenes, and scale. However, the mechanisms differ: EHS achieves effects through plasma-induced fracturing which generates strong hydraulic shock waves capable of inducing macro-fractures but carries some risk to casing integrity, while US relies on steady acoustic energy and microbubble collapse to mobilize fluids, clean pore spaces, and create micro-fractures to enhance permeability without damaging the formation.

EHS is a robust and powerful stimulation option when carefully applied, particularly for targeted fracturing. However, its higher energy release and associated risks require more cautious operation. Ultrasound stimulation, with its tunable energy delivery and safer profile, remains a strong alternative for broader applications and more sensitive formations.

Plasma Pulse Technology (PPT)

Plasma Pulse Technology (PPT) is an advanced well stimulation method that utilizes high-energy plasma discharges to generate elastic shock waves capable of penetrating deep into the reservoir. Unlike chemical or thermal methods, PPT offers a non-damaging, environmentally friendly alternative for enhancing hydrocarbon mobility and reducing near-wellbore damage. The technique involves non-linear, broadband, repetitive energy pulses that clean wellbore intervals and stimulate reservoir rock by increasing its natural resonance [20] [21].

PPT tools are conveyed via wireline or coiled tubing and positioned near the target interval. Once activated, a plasma pulse generator releases a high-voltage current (typically 3–5 kV), stored in capacitors, into a liquid medium surrounding the tool. This creates a plasma channel that expands rapidly, converting liquid to gas and generating a powerful pressure wave. These waves propagate outward as elastic energy, cleaning the perforations, removing skin damage, and inducing nano-fractures that improve oil mobility. The energy release also increases the reservoir's resonance, which has been reported to last up to 12 months and affect areas as far as 1,500 meters from the wellbore.

Operational parameters of PPT highlight its robust capabilities. It supports temperatures up to 140 °C and pressures up to 8,000 psi, significantly higher than ultrasound (US) stimulation, which operates up to 120 °C and 4,300 psi. However, PPT systems require more power—2 to 3 times higher than US—raising concerns around energy efficiency and tool robustness in long-term use. PPT tools also have a larger outer diameter (2.5 inches), while US tools are more compact (approximately 1.75 inches), making US tools better suited for narrow or complex completions.

PPT operates in a pulsed mode only, relying on accumulated energy released in short, high-intensity bursts. In contrast, US stimulation allows both continuous and pulsed operation, offering more control over energy delivery. Both methods require a liquid column to propagate waves and can be used in production and injection wells across vertical, deviated, or horizontal configurations. Conveyance methods for both include wireline, coiled tubing, or tractor-assisted deployment, depending on well orientation and depth.

One of the key differentiators between PPT and US stimulation is treatment depth. PPT has been successfully used in vertical wells up to 13,000 feet and horizontal wells as deep as 22,200 feet, slightly surpassing the maximum deployment depth of US tools, which is approximately 20,000 feet. While US stimulation focuses on precision and near-wellbore permeability enhancement, PPT extends its reach deeper into the formation and offers longer-lasting effects through resonance. Both technologies are used for permeability enhancement, formation damage removal, and the cleaning of paraffin, wax, asphaltenes, and inorganic scale. While US stimulation has demonstrated high efficiency in sand control and asphaltene removal, PPT shows strong potential for long-term fluid mobility improvements.

PPT and US stimulation are both effective tools for well stimulation, each offering distinct advantages. PPT delivers higher energy and deeper treatment effects, while US provides more controlled, energy-efficient, and adaptable interventions. One key difference is that PPT is not usually used with chemicals, but can complement chemical treatment as a pre-cleaning or post-stimulation. The choice between the two should be based on well accessibility, formation sensitivity, and desired stimulation reach.

Conclusion

Ultrasound stimulation (US) was compared in this paper against four mechanical well stimulation technologies: Dynamic Fluid Pulsing (DFP), Seismic or Low-Frequency Acoustic Stimulation, Electro-

Hydraulic Stimulation (EHS), and Plasma Pulse Technology (PPT). The comparative analysis showed that US offers a unique combination of flexibility, environmental safety, and operational efficiency. While DFP and seismic stimulation provide broader reach and long-term effects, they lack the precision and control of US, which is applied directly in the target well with minimal infrastructure. EHS and PPT deliver high-intensity pulsed energy and are effective in inducing fractures and deep reservoir activation; however, these come with increased risk to well integrity and operational complexity. US tools are smaller in diameter, suitable for a wider range of completions, can create controlled micro-fractures, applied for longer reservoir sections, and operate at competitive pressures and temperatures, all while allowing both pulsed and continuous modes. Across all comparisons, US consistently demonstrated advantages in adaptability, safety, and compatibility with other fluid stimulation techniques. While each technology presents unique strengths depending on well and reservoir conditions, US stands out as a balanced, efficient, and low-risk option. Continued research into field-scale applications and hybrid deployments may further validate its potential and expand its adoption in well stimulation practices. [Table 1](#). Below shows a summary comparison feature of US to all other stimulation tools, where it shows the clear advantages of US compared to mechanical stimulation tools. Nonetheless, there is an opportunity for the operational efficiency of ultrasound tools to be enhanced to enable stimulating longer reservoir sections in a shorter period of time.

Table 1—Summary comparison features of US to other tools.

Feature	Ultrasound (US)	DFP	Seismic	EHS	PPT
High Pressure Tolerance	✓		✓	✓	✓
High Temperature Tolerance	✓			✓	✓
Compact Tool Size	✓	✓			
Continuous & Pulsed Modes	✓				
Pulsed Only		✓	✓	✓	✓
Environmentally Friendly	✓	✓	✓		✓
Deep Reservoir Reach	✓		✓		✓
Quick Operation Time	✓	✓			
Suitable for Various Well Types	✓	✓		✓	✓
Compatible with Chemical Stimulation	✓	✓			
Risk of Damage (Casing/Cement)	Low	Low	Low	High	Moderate

References

1. Mierez, J., AlTammar, M. J., Alruwaili, K. M., & Alfaraj, R. T. (2024). Recent advances of ultrasound applications in the oil and gas industry. *Ultrasonics Sonochemistry*, **103**, 106767. <https://doi.org/10.1016/j.ultsonch.2024.106767>
2. Alfaraj, R. T., AlTammar, M. J., Mierez, J., Alruwaili, K. M., Oluyemi, G., Liu, C., Shaikh, A. A., & Alhamad, L. A. (2024, November). Laboratory assessment of high-power ultrasonic tools for well stimulation and drilling operations available to purchase. Paper presented at the International Geomechanics Conference, Kuala Lumpur, Malaysia. <https://doi.org/10.56952/IGS-2024-0218>
3. AlTammar, M. J., Alfaraj, R. T., Mierez, J., Alruwaili, K. M., Oluyemi, G., Liu, C., Shaikh, A. A., and Alomair, A. A. "Stimulation of Tight Sandstones with High-Power Ultrasound Tools in Yard-Scale Tests." Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Houston, Texas, USA, June 2025. doi: <https://doi.org/10.15530/urtec-2025-4243699>

4. Palaev AG, Shammazov IA, Dzhemilev ER. Research of the impact of ultrasonic and thermal effects on oil to reduce its viscosity. *Journal of Physics: Conference Series*. 2020;**1679**(5). doi: [10.1088/1742-6596/1679/5/052073](https://doi.org/10.1088/1742-6596/1679/5/052073).
5. Hamidi, H., Sharifi Haddad, A., Wisdom Otumudia, E., Rafati, R., Mohammadian, E., Azdarpour, A., ... Tanujaya, E. (2021). Recent applications of ultrasonic waves in improved oil recovery: A review of techniques and results. *Ultrasonics*, **110**, 106288. [28]
6. Taheri-Shakib, J., Shekarifard, A., Naderi, H., Experimental investigation of the asphaltene deposition in porous media: Accounting for the microwave and ultrasonic effects, *Journal of Petroleum Science and Engineering* (2018)
7. Vahdanikia, N., Divandari, H., Hemmati-Sarapardeh, A., Menad, N. A., Schaffie, M., & Ranjbar, M. (2020). Integrating new emerging technologies for enhanced oil recovery: Ultrasonic, microorganism, and emulsion. *Journal of Petroleum Science and Engineering*, 107229.
8. Felsenthal, M., & Ferrell, H. H. (1967). Pressure pulsing - An improved method of waterflooding fractured reservoirs. Paper presented at the SPE Permian Basin Oil Recovery Conference, Midland, Texas. <https://doi.org/10.2118/1788-MS>
9. <https://onthewavefront.com/wavefront-technology/powerwave/>
10. Davidson, B. (2015). Fluid-Pulse Technology Boosts Oil Recovery. *Journal of Petroleum Technology* **65**(12). pp 34–35. doi: [10.2118/1213-0034-JPT](https://doi.org/10.2118/1213-0034-JPT)
11. Davidson, B., Kolli, K., Spanos, T., Akar, A. A., Al Omari, M., Djellout, M. A., & Shanoun, D. (2018). Dynamic Fluid Pulsation: A Novel Approach to Reservoir Stimulation Improves Post-Stimulation Gains. SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition.
12. Tan, Y., Zhang, Y., Hui, C., Yu, C., Tian, S., Wang, T., & Wang, F. (2023). Resonance-enhanced pulsing water injection for improved oil recovery: Micromodel experiments and analysis. *Processes*, **11**(3), 957. <https://doi.org/10.3390/pr11030957>
13. Wu, F., Song, Y., Li, N., & Liu, J. (2025). Experimental investigation on enhanced oil recovery by hydraulic pulsating wave driving. *Geoenergy Science and Engineering*, **244**, 213477. <https://doi.org/10.1016/j.geoen.2024.213477>
14. Wooden, B. (2018). Seismic stimulation: An eco-friendly, effective EOR alternative. *Journal of Petroleum Technology*. <https://jpt.spe.org/seismic-stimulation-eco-friendly-effective-eor-alternative>
15. Pride, S. R., Flekkøy, E. G., & Aursjø, O. (2008). Seismic stimulation for enhanced oil recovery. *Geophysics*, **73**(5), O23–O35. <https://doi.org/10.1190/1.2968090>
16. Sun, Q., Retnanto, A., and Amani, M. (2020). Seismic vibration for improved oil recovery: A comprehensive review of literature. *International Journal of Hydrogen Energy*, **45**(29), 15151–15166. <https://doi.org/10.1016/j.ijhydene.2020.03.227>
17. Liu, S., Liu, Y., Li, H., Zhang, Q., and Lin, F. (2018). Enhanced Oil Recovery by Repetitive Electrohydraulic Shock Waves: Fracturing and Enhanced-Permeability. 2018 IEEE International Conference on High Voltage Engineering and Application (ICHVE). doi: [10.1109/ichve.2018.8642197](https://doi.org/10.1109/ichve.2018.8642197)
18. Habibi, A., Fensky, C., Perri, M., Roostaei, M., Fattahpour, V., Mahmoudi, M., Ghalambor, A., Sadrzadeh, M., and Hongbo Z. (2020). Unplugging Standalone Sand Control Screens with High-power Shock Waves: An Experimental Study. Paper presented at the SPE International Conference and Exhibition on Formation Damage Control, Lafayette, Louisiana, USA.
19. Al Harthy, A., Al Habsi, K., Al Hinai, K., and Walley, S. (2019). The First Middle East Unconventional Well Stimulation Treatment by Applying WASP Technology: Field Cases. SPE Middle East Oil and Gas Show and Conference. doi: [10.2118/195057-ms](https://doi.org/10.2118/195057-ms)

20. Perri, M. An Alternate Method of Reducing SKIN in Oil and Gas Wells. Technical Publication. BlueSpark White Papers. 52485_Reduction-skin-definition-skin.pdf
21. Patel, K., Shah, M., & Sircar, A. (2018). Plasma Pulse Technology: An uprising EOR technique. *Petroleum Research*, **3**(2), 180–188. <https://doi.org/10.1016/j.ptlrs.2018.05.001>